

# Paper Reading Notes

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## 1 A. Building Environment and IEQ

### P001 — Spatial-temporal CO2 concentration prediction in education facilities with dynamic layouts using BIM and multi-modal graph neural networks

**Authors:** Hao Liu, Kam-Ming Mark Tam, Cong Huang, Chun Yin Lai, Eric Schuldenfrei

**Major Category:** Building Environment and IEQ

**Secondary Category:** AI and Machine Learning for Buildings, BIM and Semantic Information

**Keywords:** Spatial-temporal prediction, CO2 concentration, Education facility, Space reconfigurability, BIM, Graph neural networks, Multi-modal deep learning

■**主な主張** BIM 由来の時変レイアウトと Voronoi ベースの CO2 センサ隣接グラフを multi-modal deep learning で統合し, 2 時間先予測で MAE 11.25 ppm を達成し, 比較対象中で次点の TGCN の 15.10 ppm より約 25% 低い MAE を示した. [1]

■**新規性** 静的レイアウトを前提とする既往の GNN 型 CO2 予測に対し, BIM から時変の間仕切り構成をテンソル化し, センサグラフ文脈と空間構成文脈を二枝の MMDL で統合した点に新規性がある.

■**位置付け** 教育施設の IAQ 予測を主目的としつつ, BIM 幾何情報を予測モデルの入力文脈として利用するため, Building Environment and IEQ を主分類, AI and Machine Learning for Buildings および BIM and Semantic Information を副分類として位置付ける.

### P002 — Graph-based spatial-temporal prediction and feature interaction analysis of CO2 and occupant in large indoor space

**Authors:** Cong Huang, Helen H.L. Kwok, Kwok Ho Poon, Zhaoji Wu, Fangli Hou, Jun Ma, Jack C.P. Cheng

**Major Category:** Building Environment and IEQ

**Secondary Category:** AI and Machine Learning for Buildings, Occupancy and Human-Building Interaction, Sensors, IoT and Data Integration

**Keywords:** Graph neural network, Spatial-temporal prediction, Spatial-temporal feature interactions, Indoor Air Quality, Indoor occupant, Dynamic graph learning, Occupant-in-loop

■**主な主張** 提案 AST-GCGRU は, 6 h および 12 h の multi-step prediction で graph-level R2 0.898 および 0.812 を示し, GCGRU の 0.860 および 0.699 を上回った. 12 h では occupant node R2 0.59, CO2 node R2 0.73 で, GCGRU の 0.32, 0.58 より高かった. [2]

■**新規性** 距離ベースの固定 graph ではなく, partial correlation で初期化した graph を ST-Att で学習中に更新し, time-lag nodes により 30 min CO2 sensor と 6 min occupant sensor の異なる時間解像度を保持したまま, CO2 と occupancy を同一 graph で同時予測・interaction analysis する構成を提示した.

■**位置付け** 大空間における IAQ と occupancy の同時予測・相互作用分析を主目的とするため, Building Environment and IEQ を主分類とし, AI and Machine Learning for Buildings, Occupancy and Human-Building Interaction, Sensors, IoT and Data Integration を副分類として位置付ける.

## P005 — CO2 concentration prediction in office spaces using physics-informed neural network based on number of occupants and IoT sensor data

**Authors:** Jehyun Kim, Gihoon Kim, Jong-Il Bang, Anseop Choi, Minki Sung

**Major Category:** Building Environment and IEQ

**Secondary Category:** AI and Machine Learning for Buildings, Occupancy and Human-Building Interaction, Sensors, IoT and Data Integration

**Keywords:** CO2 prediction, Physics-informed neural network, Occupancy, HVAC operation, IoT, Indoor air quality

■**主な主張** occupancy と HVAC/IoT data に CO2 mass-balance 制約を加えた PINN で office CO2 を予測した. [3]

■**新規性** data-driven CO2 prediction へ mass-balance equation を physics-informed loss として導入した.

■**位置付け** CO2 prediction を中心とする IEQ × PINN 研究として位置付ける.

## P006 — Understanding the effect of outdoor pollution episodes and HVAC type on indoor air quality

**Authors:** Tristalee Mangin, Zachary Barrett, Zachary Palmer, Dillon Tang, Sean Nielson, Darrah Sleeth, Kerry Kelly

**Major Category:** Building Environment and IEQ

**Secondary Category:** HVAC and Building Services, Sensors, IoT and Data Integration

**Keywords:** PM2.5, HVAC, Infiltration, Wildfire smoke, Indoor air quality, Low-cost sensors

■**主な主張** 18 か月, 21 台の PM2.5 sensor で outdoor pollution episode と HVAC type による室内 PM2.5 infiltration 差を評価した. [4]

■**新規性** inversion, dust, wildfire smoke を HVAC type 別に long-term 比較した.

■**位置付け** outdoor-to-indoor PM2.5 infiltration と HVAC 影響の実測研究として位置付ける.

## P014 — CO2- and PM2.5-Focused Optimal Ventilation Strategy Based on Predictive Control

**Authors:** Young Jae Choi, Eun Ji Choi, Jae Yoon Byun, Hyeun Jun Moon, Min Ki Sung, Jin Woo Moon

**Major Category:** Building Environment and IEQ

**Secondary Category:** HVAC and Building Services, AI and Machine Learning for Buildings

**Keywords:** CO2, PM2.5, Ventilation, Predictive control, VAV, Indoor air quality, Co-simulation

■**主な主張** CO2/PM2.5 予測を用いる VAV ventilation control を提案し, co-simulation で両汚染物質の制御性能を評価した. [5]

■**新規性** CO2 と PM2.5 の予測 error に応じ control priority を動的に切替えた.

■位置付け multi-pollutant IAQ predictive ventilation control 研究として位置付ける.

## 2 B. HVAC and Building Services

### P015 — Deep reinforcement learning control for co-optimizing energy consumption, thermal comfort, and indoor air quality in an office building

**Authors:** Fangzhou Guo, Sang woo Ham, Donghun Kim, Hyeun Jun Moon

**Major Category:** HVAC and Building Services

**Secondary Category:** AI and Machine Learning for Buildings, Building Environment and IEQ, Building Energy and ZEB

**Keywords:** Deep reinforcement learning, HVAC control, Energy, Thermal comfort, CO<sub>2</sub>, PM<sub>2.5</sub>, DDPG

■**主な主張** DDPG-based DRL で HVAC を制御し, energy, thermal comfort, CO<sub>2</sub>, PM<sub>2.5</sub> を同時最適化した. [6]

■**新規性** 複数 HVAC actuator と IEQ-energy trade-off を単一 DRL supervisory controller で扱った.

■**位置付け** HVAC supervisory control における DRL-based multi-objective optimization 研究として位置付ける.

### 3 C. Building Energy and ZEB

## 4 D. BMS, BAS, BACS and Building Automation

### P011 — Integrating the Brick Schema and Control Description Language for automation and controls in BIM

**Authors:** Esben Visby Fjerbæk, Kevin Michael Smith, Christian Anker Hviid

**Major Category:** BMS, BAS, BACS and Building Automation

**Secondary Category:** BIM and Semantic Information, HVAC and Building Services

**Keywords:** BACS, Brick Schema, Control Description Language, BIM, HVAC controls, Interoperability

■**主な主張** Brick Schema と CDL を統合し, BIM 由来 HVAC equipment と control logic を machine-readable knowledge graph で接続した. [7]

■**新規性** BACS context と control sequence を Brick+CDL で一体表現した.

■**位置付け** BACS/HVAC control semantics と BIM interoperability 研究として位置付ける.

## 5 E. BIM and Semantic Information

### P009 — A systematic comparison and evaluation of building ontologies for deploying data-driven analytics in smart buildings

**Authors:** Zhangcheng Qiang, Stuart Hands, Kerry Taylor, Subbu Sethuvenkatraman, Daniel Hugo, Pouya Ghiasnezhad Omran, Madhawa Perera, Armin Haller

**Major Category:** BIM and Semantic Information

**Secondary Category:** Reviews, Frameworks and Standards, Sensors, IoT and Data Integration

**Keywords:** Building ontology, Data interoperability, Brick Schema, RealEstateCore, Project Haystack, Digital Buildings, Ontology evaluation

■主な主張 Brick, RealEstateCore, Project Haystack, Digital Buildings の 4 ontology を TBox/ABox 両面で比較し, alignment の必要性を示した. [8]

■新規性 axiomatic quality と実建物 data による completeness/expressiveness を同一 framework で評価した.

■位置付け smart-building ontology interoperability の比較評価研究として位置付ける.

### P010 — BIM-to-BRICK: Using graph modeling for IoT/BMS and spatial semantic data interoperability within digital data models of buildings

**Authors:** Filippo Vittori, Chuan Fu Tan, Anna Laura Pisello, Adrian Chong, Cristina Piselli, Clayton Miller

**Major Category:** BIM and Semantic Information

**Secondary Category:** BMS, BAS, BACS and Building Automation, Sensors, IoT and Data Integration, Digital Twin

**Keywords:** BIM, Brick Schema, BMS, IoT, Knowledge graph, Data interoperability, Digital Twin

■主な主張 BIM-to-BRICK により BIM 情報を Brick knowledge graph へ自動変換し, BIM と BMS/IoT data の semantic interoperability を実装した. [9]

■新規性 BIM-to-Brick mapping を graph modeling で自動化した.

■位置付け BIM-BMS/IoT 間 semantic interoperability 研究として位置付ける.

### P016 — Brick: Metadata schema for portable smart building applications

**Authors:** Bharathan Balaji, Arka Bhattacharya, Gabriel Fierro, Jingkun Gao, Joshua Gluck, Dezhi Hong, Aslak Johansen, Jason Koh, Joern Ploennigs, Yuvraj Agarwal, Mario Bergés, David Culler, Rajesh K. Gupta, Mikkel Baun Kjærgaard, Mani Srivastava, Kamin Whitehouse

**Major Category:** BIM and Semantic Information

**Secondary Category:** BMS, BAS, BACS and Building Automation, Sensors, IoT and Data Integration

**Keywords:** Brick Schema, Building metadata, Semantic model, RDF, SPARQL, Portable applications

■主な主張 要検討 [10]

■新規性 要検討

■位置付け 要検討

## P017 — BOT: The Building Topology Ontology of the W3C Linked Building Data Group

**Authors:** Mads Holten Rasmussen, Maxime Lefrançois, Georg Ferdinand Schneider, Pieter Pauwels

**Major Category:** BIM and Semantic Information

**Secondary Category:** Reviews, Frameworks and Standards

**Keywords:** BOT, Building topology ontology, Linked Building Data, Ontology, Semantic Web, RDF

■主な主張 要検討 [11]

■新規性 要検討

■位置付け 要検討

## P018 — Semantic model-based large-scale deployment of AI-driven building management applications

**Authors:** Kan Xu, Zhe Chen, Fu Xiao, Jing Zhang, Hanbei Zhang, Tianyou Ma

**Major Category:** BIM and Semantic Information

**Secondary Category:** AI and Machine Learning for Buildings, BMS, BAS, BACS and Building Automation

**Keywords:** Semantic model, Brick Schema, AI applications, Building management, Data interoperability, Large-scale deployment

■主な主張 要検討 [12]

■新規性 要検討

■位置付け 要検討

## P019 — A framework for integrating BIM, building automation systems, and IoT: an application in design and operation using open standards

**Authors:** Shu Tang, Dennis R. Shelden, Xinghua Gao, Cheng Zhang, Jianli Hao, Fangyu Guo, Haoyu Zhang

**Major Category:** BIM and Semantic Information

**Secondary Category:** BMS, BAS, BACS and Building Automation, Sensors, IoT and Data Integration

**Keywords:** BIM, BAS, IoT, BOT, Brick Schema, Semantic Web, Open standards

■主な主張 要検討 [13]

■新規性 要検討

■位置付け 要検討

## P021 — Metadata schema for virtual building models in digital twins: VB schema implemented in GPT-based applications

**Authors:** Jeyoon Lee, Sungmin Yoon

**Major Category:** BIM and Semantic Information

**Secondary Category:** Digital Twin, Retrieval, Query, LLM and Agent

**Keywords:** VB schema, Virtual building models, Digital Twin, Ontology, Brick Schema, GPT, HVAC

■主な主張 要検討 [14]

■新規性 要検討

■位置付け 要検討

## P023 — Metadata schema for virtual sensors in digital twin-enabled building systems using Brick schema

**Authors:** Syed Mostasim Hasnain Saif, Jeyoon Lee, Peng Wang, Sungmin Yoon

**Major Category:** BIM and Semantic Information

**Secondary Category:** Digital Twin, Sensors, IoT and Data Integration

**Keywords:** Virtual sensors, Brick Schema, Metadata schema, Digital Twin, Ontology, Building systems

■主な主張 要検討 [15]

■新規性 要検討

■位置付け 要検討

## P028 — Semantic Web Technologies for Indoor Environmental Quality: A Review and Ontology Design

**Authors:** Alex Donkers, Dujuan Yang, Bauke de Vries, Nico Baken

**Major Category:** BIM and Semantic Information

**Secondary Category:** Building Environment and IEQ, Reviews, Frameworks and Standards

**Keywords:** Semantic web, Linked data, Ontology, Indoor environmental quality, Building Performance Ontology, BIM, IoT

■主な主張 IEQ modellingに必要なBIM/IoT由来の異種情報を扱うため、building topologyとstatic/dynamic propertiesを統合するBuilding Performance Ontologyを構築し、複数levels of detailを備えた拡張可能なsemantic data modelを提示した。 [16]

■新規性 既存 ontology の systematic review から共通 design patterns を抽出し, topology, static properties, observations, actuations を lightweight upper ontology と lower-level ontologies へ統合した点に新規性がある.

■位置付け IEQ 向け semantic integration の基盤研究であり, BIM, IoT, BMS の異種情報を building context と結び付ける ontology 設計として位置付ける.

## 6 F. Digital Twin

### P004 — How to measure and control indoor air quality based on intelligent digital twin platforms: A case study in China

**Authors:** Yuchong Qian, Jiawei Leng, Kai Zhou, Yuxuan Liu

**Major Category:** Digital Twin

**Secondary Category:** Building Environment and IEQ, BIM and Semantic Information, Sensors, IoT and Data Integration

**Keywords:** Digital Twin, Indoor air quality, BIM, IoT, Machine learning, Monitoring and control

■**主な主張** IoT, BIM, machine learning を統合した IAQ 用 digital twin platform を構築し, monitoring, prediction, visualization, control を実装した. [17]

■**新規性** IAQ の Sense-Integrate-Retrieve-Analyze-Feedback を単一 DT platform へ統合した.

■**位置付け** BIM-IoT-ML 統合による IAQ Digital Twin platform 研究として位置付ける.

### P007 — A BIM-enabled digital twin framework for real-time indoor environment monitoring and visualization by integrating autonomous robotics, LiDAR-based 3D mobile mapping, IoT sensing, and indoor positioning technologies

**Authors:** Xi Hu, Rayan H. Assaad

**Major Category:** Digital Twin

**Secondary Category:** Building Environment and IEQ, BIM and Semantic Information, Sensors, IoT and Data Integration

**Keywords:** Digital Twin, BIM, IoT, Indoor environmental monitoring, Indoor positioning, Robotics, LiDAR

■**主な主張** BIM, IoT, robotics, LiDAR, UWB positioning を統合し, real-time indoor environment monitoring を DT 上で可視化した. [18]

■**新規性** mobile robotic sensing と indoor positioning を BIM-enabled DT へ統合した.

■**位置付け** BIM-enabled spatial Digital Twin による IEQ monitoring 研究として位置付ける.

### P022 — Ontology-enabled AI agent-driven intelligent digital twins for building operations and maintenance

**Authors:** Sungmin Yoon, Jihwan Song, Jiteng Li

**Major Category:** Digital Twin

**Secondary Category:** Retrieval, Query, LLM and Agent, BIM and Semantic Information, Fault Detection, Diagnostics and Maintenance

**Keywords:** Digital Twin, AI agent, Ontology, Building operations, Maintenance, OpenAI API, LangChain

■主な主張 要検討 [19]

■新規性 要検討

■位置付け 要検討

## P026 — Modeling indoor thermal comfort in buildings using digital twin and machine learning

**Authors:** Ziad ElArwady, Ahmed Kandil, Mohanad Afiffy, Mohamed Marzouk

**Major Category:** Digital Twin

**Secondary Category:** Building Environment and IEQ, AI and Machine Learning for Buildings, BIM and Semantic Information

**Keywords:** Digital Twin, Thermal comfort, Machine learning, BIM, IoT, Indoor environment

■主な主張 要検討 [20]

■新規性 要検討

■位置付け 要検討

## 7 G. Sensors, IoT and Data Integration

### P013 — Smart metering data enhancement in sustainable buildings via knowledge graph-guided graph neural networks

**Authors:** Fang He, Jiaqi Fan, Yang Deng, Xiaoyang Zhang, Ka Tai Lau, Dan Wang

**Major Category:** Sensors, IoT and Data Integration

**Secondary Category:** AI and Machine Learning for Buildings, BIM and Semantic Information

**Keywords:** Smart metering, Building knowledge graph, Graph neural network, Data enhancement, Knowledge embedding, Brick Schema

■**主な主張** building knowledge graph と GNN を融合した BuildKnow により smart-meter data enhancement を行い, 3 実建物で評価した. [21]

■**新規性** knowledge graph 由来の meter dependence を GNN の学習 prior として利用した.

■**位置付け** smart-meter data quality と knowledge-guided GNN 研究として位置付ける.

## 8 H. Fault Detection, Diagnostics and Maintenance

## 9 I. AI and Machine Learning for Buildings

## 10 J. Retrieval, Query, LLM and Agent

### P012 — An ontology-aided, natural language-based approach for multi-constraint BIM model querying

**Authors:** Mengtian Yin, Llewellyn Tang, Chris Webster, Shen Xu, Xiongyi Li, Huaquan Ying

**Major Category:** Retrieval, Query, LLM and Agent

**Secondary Category:** BIM and Semantic Information

**Keywords:** Natural language query, BIM retrieval, Ontology, Semantic parser, SPARQL, IFC, Multi-constraint query

■**主な主張** ontology-aided semantic parser で multi-constraint natural language query を IFC-compliant SPARQL へ変換し, 225 NLQs で 91.11% の overall accuracy を報告した. [22]

■**新規性** 複合 NL query を ontology を介して structured BIM query へ変換した.

■**位置付け** 自然言語による BIM retrieval/query 研究として位置付ける.

## 11 K. Occupancy and Human-Building Interaction

## 12 L. Reviews, Frameworks and Standards

### P003 — Digital twin applications for indoor environmental quality in healthy building research: a systematic review of indicators, methods and computational paradigms

**Authors:** Alessandro D’Amico, Edoardo Currà, Philomena M. Bluysen

**Major Category:** Reviews, Frameworks and Standards

**Secondary Category:** Digital Twin, Building Environment and IEQ, BIM and Semantic Information

**Keywords:** Digital Twin, Indoor environmental quality, Healthy buildings, BIM, IoT, Systematic review

■**主な主張** 62 件の DT × IEQ 研究を systematic review し, building semantics と occupant-related data の統合不足を整理した. [23]

■**新規性** DT × IEQ を指標, 方法, computational paradigm の観点から横断整理した.

■**位置付け** DT × IEQ の研究ギャップを整理する systematic review として位置付ける.

### P008 — Digital twin-enabled smart facility management: A bibliometric review

**Authors:** Obaidullah Hakimi, Hexu Liu, Osama Abudayyeh

**Major Category:** Reviews, Frameworks and Standards

**Secondary Category:** Digital Twin, BIM and Semantic Information, Fault Detection, Diagnostics and Maintenance

**Keywords:** Digital Twin, Facility management, BIM, Semantic interoperability, Predictive maintenance, Autonomous control

■**主な主張** 248 件を bibliometric review し, DT-enabled FM の主要潮流と semantic interoperability, predictive maintenance 等の将来課題を整理した. [24]

■**新規性** DT-enabled FM 研究の構造と future needs を bibliometric に体系化した.

■**位置付け** Digital Twin × facility management の bibliometric review として位置付ける.

### P020 — Ontology-driven energy management in smart buildings: A comprehensive review of methodologies, tools, and challenges

**Authors:** Adamantios Bampoulas, Kaiwalya Raj, Anandhi Parthiban, Mohammad Saffari, Eleni Mangina

**Major Category:** Reviews, Frameworks and Standards

**Secondary Category:** BIM and Semantic Information, Building Energy and ZEB

**Keywords:** Ontology, Building energy systems, Semantic interoperability, Knowledge representation, Brick, SAREF, Digital Twin

■**主な主張** 81 件の primary study を対象に ontology-driven BES を横断整理し, ontology は standardised data flows と multi-agent coordination を支援する一方, domain coverage, emerging industry standards との alignment, predictive modelling 向け time-series integration にギャップが残ると示した. [25]

■**新規性** occupant behaviour, IoT sensor data, demand response 等に限定されていた既往 review に対し, DT integration, occupant-centric control, semantic interoperability, AI-driven optimisation を含む BES 全体の統合的 review を提示した.

■**位置付け** ontology と AI/ML, Digital Twin, real-time energy management の接続を俯瞰する review であり, semantic model を prediction/control へ接続する研究課題の上位背景として位置付ける.

## **P024 — Commercial building indoor environmental quality models: A critical review**

**Authors:** Soheil Roumi, Fan Zhang, Rodney A. Stewart, Mattheos Santamouris

**Major Category:** Reviews, Frameworks and Standards

**Secondary Category:** Building Environment and IEQ

**Keywords:** Indoor environmental quality, IEQ models, Commercial buildings, Critical review, Thermal comfort, Indoor air quality

■**主な主張** 要検討 [26]

■**新規性** 要検討

■**位置付け** 要検討

## **P025 — Machine Learning and Deep Learning Methods for Enhancing Building Energy Efficiency and Indoor Environmental Quality - A Review**

**Authors:** Paige Wenbin Tien, Shuangyu Wei, Jo Darkwa, Christopher Wood, John Kaiser Calautit

**Major Category:** Reviews, Frameworks and Standards

**Secondary Category:** AI and Machine Learning for Buildings, Building Environment and IEQ, Building Energy and ZEB

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■**主な主張** 要検討 [27]

■**新規性** 要検討

■**位置付け** 要検討

## P027 — Developing an integrative framework for digital twin applications in the building construction industry: A systematic literature review

**Authors:** Wuyan Long, Zhikang Bao, Ke Chen, S. Thomas Ng, Ibrahim Yahaya Wuni

**Major Category:** Reviews, Frameworks and Standards

**Secondary Category:** Digital Twin

**Keywords:** Digital Twin, Building construction, Systematic literature review, Integrative framework, Construction, Operation and maintenance

■主な主張 要検討 [28]

■新規性 要検討

■位置付け 要検討

13 Uncategorized

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